De La Salle University

Borrowed Stability: Decomposing Philippine Peso Depreciation Using DXY Normalization and ARIMA Counterfactual Analysis, January 2025 to March 2026

Department of Financial Management

De La Salle University, Manila, Philippines

By:

Aaron Joshua E. Sison

Undergraduate Student

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Author Note

This paper was prepared as part of a quantitative finance research exercise. Correspondence concerning this paper should be addressed to the author. No conflicts of interest are declared. Data and code are available from the author upon request. Exchange rate data were sourced from Yahoo Finance (Yahoo Finance, 2026a, 2026b). Analysis was performed in Python 3.12 using statsmodels (Seabold & Perktold, 2010) and matplotlib (Hunter, 2007).

Abstract

The Philippine peso presents a misleading image of stability. Since Donald Trump's inauguration on January 20, 2025, the USD/PHP spot rate rose from 58.15 to 60.21, a nominal depreciation of 3.54%. During the same period, the US Dollar Index (DXY) declined from 108.06 to 100.15, a deterioration of 7.32% in trade-weighted dollar strength. This paper decomposes the observed exchange rate movement into two components: peso-specific depreciation and USD-specific weakness. Two counterfactual methods are applied. The first is a DXY-normalization procedure that adjusts the actual USDPHP rate by the ratio of the DXY at inauguration to its current level, producing a stable-dollar counterfactual rate. The second is an ARIMA(1,1,1) model with drift, fitted on pre-inauguration data from July 2020 to January 2025, and projected forward through March 2026. The DXY-normalized rate for March 27, 2026 is 64.93 PHP/USD, implying a fundamental PHP depreciation of 11.66% against a stable dollar. The ARIMA forecast yields 60.78 PHP/USD, broadly consistent with the nominal rate and the pre-existing depreciation trend. The difference between the normalized counterfactual and the actual rate constitutes a PHP cushion of 4.72 PHP/USD, representing the portion of peso strength attributable to dollar weakness rather than peso fundamentals. A hypothesized stable-dollar rate of approximately 70 PHP/USD is directionally correct but overstated by approximately 5 PHP/USD, equivalent to a DXY decline of 13% rather than the observed 7.32%. The results indicate that the peso is functionally weaker than exchange boards report, and that BSP reserve operations and OFW remittance flows provide additional suppression not captured by either model.

Keywords: Philippine peso, exchange rate decomposition, DXY normalization, ARIMA, counterfactual analysis, USD weakness, BSP

Introduction

The Philippine peso occupies an uncomfortable position in Southeast Asian currency markets. Its exchange rate reflects three compounding forces at once: domestic monetary policy, global dollar dynamics, and the structural peculiarity of an economy whose external accounts depend heavily on remittance inflows from approximately 10.6 million overseas Filipino workers (Bangko Sentral ng Pilipinas [Bangko Sentral ng Pilipinas], 2024). The resulting rate is a composite signal. Separating its components requires an explicit analytical framework.

This paper addresses one specific decomposition question. Since January 20, 2025, the US dollar has weakened materially against a trade-weighted basket of major currencies. The DXY, maintained by ICE Benchmark Administration (ICE Benchmark Administration, 2023), declined 7.32% over this period. The USDPHP rate rose by only 3.54%. The arithmetic is consequential: a peso that appears to depreciate modestly against a falling dollar is, in fundamental terms, depreciating more substantially than the quoted rate implies. The dollar's weakness is providing cover. This paper quantifies the magnitude of that cover.

Two analytical instruments are deployed. The first is a DXY-normalization formula that recalculates the USDPHP rate under a counterfactual of stable dollar strength. The second is an ARIMA(1,1,1) model fitted on pre-Trump data, projecting forward what PHP depreciation would look like absent the structural break of the new administration's trade and fiscal policies. The combination of these approaches permits a triangulated estimate of the peso's fundamental position.

1 Objectives

Five objectives guide this analysis. First, to document and quantify the actual USDPHP movement from inauguration day through March 27, 2026. Second, to construct a

DXY-normalization counterfactual that strips USD weakness from the observed exchange rate. Third, to develop an ARIMA(1,1,1) forecast extending pre-Trump PHP depreciation trends into the post-inauguration period. Fourth, to evaluate the quantitative plausibility of a 70 PHP/USD stable-dollar hypothesis. Fifth, to calculate the PHP cushion at each trading day of the post-inauguration period and characterize its trajectory.

2 Literature Review

2.1 Exchange Rate Determination Theory

The monetary approach to exchange rate determination holds that bilateral rates reflect relative money supply growth, inflation differentials, and interest rate parity conditions (Frenkel, 1976). Dornbusch (1976) extended this framework with the overshooting model, demonstrating that exchange rates respond to monetary shocks in ways that temporarily exceed long-run equilibrium adjustments. The Dornbusch (1976) model's core insight is relevant here: exogenous shocks to the dollar, such as those induced by tariff announcements and fiscal uncertainty under the Trump administration, can produce immediate exchange rate movements that exceed what fundamental values justify.

The most consequential empirical finding in exchange rate economics remains Meese and Rogoff's (1983) demonstration that structural FX models fail to outperform a random walk in out-of-sample forecasting. This result motivates the use of ARIMA models in this analysis. A random walk with drift is the methodologically honest benchmark for exchange rate projection: it imposes minimal structural assumptions and produces forecasts that are not demonstrably worse than those from elaborate causal models. The ARIMA(1,1,1) specification generalizes the random walk by allowing first-order autocorrelation in daily changes, capturing short-run momentum while retaining the integrated-of-order-one characterization that characterizes exchange rate levels.

Rogoff's (1996) survey of the purchasing power parity puzzle documents the slow

convergence of exchange rates to fundamental equilibrium, typically spanning three to five years. This finding cautions against expecting the PHP cushion identified in this analysis to dissipate quickly. Deviations from DXY-implied equilibrium can persist substantially longer than models suggest.

2.2 Philippine Peso in Macroeconomic Context

The Bangko Sentral ng Pilipinas operates under a formal inflation-targeting framework while maintaining a managed float exchange rate regime. Guinigundo (2014) describes the BSP's use of FX intervention as a smoothing mechanism rather than a peg, with reserves deployed to moderate excessive peso volatility rather than to defend a specific rate. This policy posture implies that observed USDPHP rates are policy- conditioned: the BSP systematically suppresses movements that market forces alone would produce.

Calvo and Reinhart (2002) coined the phrase "fear of floating" to describe emerging market central banks that formally adopt flexible exchange rate regimes but intervene so frequently that outcomes resemble a soft peg. The BSP's reserve deployment patterns are consistent with this characterization. The practical consequence for this analysis is that both the actual USDPHP rate and the DXY-normalized counterfactual likely understate the peso's fundamental weakness: the normalization removes the dollar component but does not remove the BSP intervention component.

OFW remittances constitute approximately 8.2% of Philippine GDP and provide a persistent structural inflow of foreign exchange (World Bank, 2023). During periods of dollar weakness, these remittances deliver fewer pesos per dollar to recipient households, creating a domestic welfare effect that the quoted exchange rate obscures. The remittance channel also limits the extent to which USDPHP can rise, as the combination of BSP management and structural dollar supply creates a natural ceiling on peso depreciation.

Ilzetzki et al. (2019) classify the Philippine peso's de facto exchange rate arrange-

ment as a managed float with dollar anchor tendencies, meaning the BSP's intervention function responds asymmetrically to depreciation pressure versus appreciation pressure. This asymmetry is directly relevant to the counterfactual constructed here, as it implies the peso's path under a stable dollar would have been influenced by BSP decisions in addition to market fundamentals.

2.3 The US Dollar Index and Trade-Weighted Measures

The DXY, first introduced by the Federal Reserve Board and subsequently administered by ICE Benchmark Administration, measures USD strength against a basket of six currencies: the euro (57.6%), Japanese yen (13.6%), British pound (11.9%), Canadian dollar (9.1%), Swedish krona (4.2%), and Swiss franc (3.6%) (ICE Benchmark Administration, 2023). The basket's heavy euro weighting means DXY movements are substantially influenced by EUR/USD dynamics, which may differ from PHP-specific dollar demand patterns.

This compositional limitation is acknowledged as a source of approximation error in the normalization formula. The PHP does not appear in the DXY basket, and the Philippine trade exposure to DXY-basket currencies differs from the implied basket weights. Nevertheless, the DXY captures aggregate dollar momentum with sufficient fidelity to serve as a first-order control for exogenous USD movements, given the dollar's role as the primary invoicing currency for Philippine trade and the denomination currency for OFW remittances.

Fajgelbaum and Khandelwal (2022) document the relationship between US tariff policy and trade flows, demonstrating that broad-based tariff imposition disrupts established supply chains and generates exchange rate volatility in affected bilateral pairs. The Trump administration's tariff announcements beginning in early 2025 constitute the exogenous policy shock that partially accounts for the DXY's decline over the study period. Reduced confidence in dollar-denominated assets, combined with concerns over US fiscal

sustainability, contributed to the sustained DXY deterioration observed in the data (International Monetary Fund, 2025).

2.4 ARIMA in Exchange Rate Forecasting

Hyndman and Athanasopoulos (2021) provide the definitive contemporary treatment of ARIMA modeling, defining the ARIMA(p, d, q) specification where p is the autoregressive order, d is the degree of differencing, and q is the moving average order. Box et al. (2015) established the iterative identification-estimation-diagnostic cycle that remains the methodological standard.

For exchange rate series, ARIMA$(1, 1, 1)$ with drift is a parsimonious specification that acknowledges non-stationarity at the level, induces stationarity through first differencing, captures first-order momentum in daily changes, and incorporates a drift parameter to represent secular depreciation trends. The model makes no causal claim about the sources of PHP movement. It characterizes the historical process and projects it forward, providing a trend-consistent counterfactual against which post-inauguration behavior can be compared.

3 Methodology

3.1 Data Sources and Collection

Daily closing prices for two financial instruments were retrieved from Yahoo Finance via the `yfinance` Python library (version 0.2.50) (Yahoo Finance, 2026a, 2026b). The instruments are `USDPHP=X` (USD to PHP spot rate, sourced from the over-the-counter interbank market as aggregated by Yahoo Finance) and `DX-Y.NYB` (US Dollar Index continuous front-month contract, sourced from ICE Futures via NYSE Arca). The data collection window spans January 2, 2018 to March 27, 2026. After alignment on common trading days and removal of missing observations through list-wise deletion, the merged

dataset contains 2,070 daily observations.

The January 20, 2025 inauguration of Donald Trump as the 47th US President serves as the structural break point. All observations prior to this date constitute the pre-inauguration (training) period. All observations on or after this date constitute the post-inauguration (evaluation) period, comprising 297 trading days through March 27, 2026.

The training window for the ARIMA model begins July 1, 2020, not January 2018. This choice excludes the acute COVID-19 disruption of currency markets, during which USDPHP exhibited non-representative volatility attributable to pandemic capital flows rather than structural PHP dynamics. The training sample thus contains 1,145 pre- inauguration observations. Figure 2 displays the full time series of USDPHP and DXY from 2018 through the analysis date.

3.2 DXY Normalization Framework

The DXY-normalization counterfactual addresses the following question: what would the USDPHP rate be if the US Dollar Index had remained constant at its January 20, 2025 level? The framework rests on a proportionality assumption. If the DXY at time t is D_t and the DXY at the inauguration baseline is D_0, then the dollar has weakened by the factor $k_t = D_0/D_t$ relative to the basket. Under proportionality, a peso rate that reflects PHP-specific fundamentals alone, isolated from USD-side effects, is given by:

$$\hat{e}_t = e_t \cdot \frac{D_0}{D_t} \tag{1}$$

where e_t denotes the actual USDPHP spot rate at time t, $D_0 = 108.06$ is the DXY at inauguration, D_t is the contemporaneous DXY value, and $\hat{e}_t$ is the DXY-normalized counterfactual rate. When the DXY falls below D_0, the adjustment factor D_0/D_t exceeds unity, producing $\hat{e}_t > e_t$. This means the peso is functionally weaker than the quoted rate implies.

The PHP cushion at each trading day is defined as the signed differential:

$$\Gamma_t = \hat{e}_t - e_t \tag{2}$$

A positive Γ_t indicates that USD weakness is masking PHP depreciation. The quoted rate understates the PHP's fundamental deterioration by Γ_t pesos per dollar. Figure 4 displays the trajectory of Γ_t across the post-inauguration period.

The proportionality assumption in Equation 1 has three notable limitations. First, the DXY basket excludes the Philippine peso, meaning the basket-implied dollar movement may not correspond precisely to USD-PHP bilateral dynamics. Second, the assumption imposes a linear scaling that ignores potential non-linearities in the PHP's sensitivity to DXY movements. Third, BSP intervention creates policy-induced deviations that neither the actual rate nor the counterfactual captures in full. These limitations counsel treating the DXY-adjusted rate as a directional indicator rather than a precise measurement.

3.3 ARIMA(1,1,1) Counterfactual Model

The second counterfactual approach employs a univariate ARIMA model to project the pre-inauguration PHP depreciation trend into the post-inauguration period. The model specification is ARIMA$(1, 1, 1)$ with drift, defined as:

$$\nabla y_t = c + \phi_1 \nabla y_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1}, \qquad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2) \tag{3}$$

where $\nabla y_t = y_t - y_{t-1}$ is the first difference of the daily USDPHP closing price, c is a constant capturing the secular depreciation drift, ϕ_1 is the first-order autoregressive coefficient on the lagged first difference, θ_1 is the first-order moving average coefficient applied to the lagged residual ε_{t-1}, and ε_t is a white noise innovation process.

The choice of $d = 1$ follows from the established non-stationarity of exchange rate levels in the empirical literature (Meese & Rogoff, 1983). Exchange rates are integrated of

order one; their levels contain a stochastic trend, but their first differences are stationary. The orders $p = 1$ and $q = 1$ are selected on the basis of information criterion parsimony. Estimation of the model on the 1,145- observation training sample yielded an Akaike Information Criterion of -19.16 and a Bayesian Information Criterion of 1.01.

Maximum likelihood estimation was performed using the ARIMA implementation in the `statsmodels` Python library (Seabold & Perktold, 2010). Post-inauguration forecasts were generated for all 297 trading days from January 20, 2025 to March 27, 2026, with 90% forecast confidence intervals derived from the forecast error variance-covariance matrix.

The ARIMA counterfactual answers a distinct question from the DXY normalization. Rather than asking what USDPHP would be given a stable dollar, it asks what USDPHP would be had the pre-inauguration trend continued undisturbed. These are complementary but non-identical questions. The DXY normalization holds the dollar constant and lets PHP fundamentals move; the ARIMA projection holds the structural process constant and projects forward.

3.4 Computation Environment

All analyses were performed in Python 3.12 using the following libraries: yfinance 0.2.50 for data retrieval (Yahoo Finance, 2026a, 2026b), statsmodels 0.14 for ARIMA estimation (Seabold & Perktold, 2010), pandas 2.2 and numpy 1.26 for data manipulation, and matplotlib 3.9 for visualization (Hunter, 2007). Package management and environment isolation were handled through the `uv` runtime manager. All computations are fully reproducible and available from the author upon request.

3.5 Scope

This analysis is bounded along four dimensions. The geographic scope is the Philippine interbank foreign exchange market, with the USDPHP rate defined as the over-the-

counter mid-rate aggregated by Yahoo Finance from dealer quotes. The temporal scope spans January 2, 2018 to March 27, 2026 for the full data window, with the structural evaluation period restricted to January 20, 2025 through March 27, 2026. The instruments in scope are limited to two: the `USDPHP=X` spot rate and the `DX-Y.NYB` Dollar Index continuous contract. All other currency pairs, Philippine equity or bond markets, and macroeconomic time series fall outside the scope of this study.

The methodological scope is explicitly narrow. Two counterfactual methods are applied: a DXY-proportionality normalization and a univariate ARIMA(1,1,1) model. This paper does not construct a structural VAR, a gravity model of bilateral trade flows, or a general equilibrium framework. It does not forecast the peso's future path. It documents a historical decomposition of an observed rate across a fixed window.

3.6 Assumptions

Six operational assumptions underlie the quantitative framework.

The first assumption is DXY proportionality. The normalization in Equation 1 assumes that the USDPHP rate responds proportionally to changes in the DXY index. A one percent decline in DXY translates directly to a proportional upward adjustment in the counterfactual USDPHP rate. This is a first-order linear approximation. Non-linearities in the PHP-DXY relationship, which may arise from threshold effects or portfolio rebalancing dynamics, are not modeled.

The second assumption is inauguration-day as a clean structural break. January 20, 2025 is treated as the single inflection point separating the training and evaluation periods. Market anticipation of the Trump administration's policies, which began pricing in during the November 2024 election campaign, is not modeled as a pre-break contamination. This simplification introduces minor estimation bias in the ARIMA training window.

The third assumption is I(1) stationarity of USDPHP levels. The ARIMA specification with $d = 1$ assumes the rate contains exactly one unit root. Neither a formal

Augmented Dickey-Fuller test nor a KPSS test is reported in this paper. The assumption is grounded in the established empirical consensus (Meese & Rogoff, 1983) rather than sample-specific testing.

The fourth assumption is training sample representativeness. The ARIMA model is fitted on July 2020 to January 2025 data. This window excludes the acute COVID-19 disruption of early 2020 but includes the post-pandemic recovery, the 2022 global rate hike cycle, and the 2023–2024 dollar strengthening period. The training window is treated as representative of the structural PHP depreciation process absent the Trump-era policy shock.

The fifth assumption is data integrity. Yahoo Finance closing prices are treated as sufficiently accurate proxies for interbank mid-rates. Bid-ask spreads, settlement date conventions, and time-zone alignment between the Philippine interbank market and the DXY futures contract are not adjusted for.

The sixth assumption is BSP non-peg. The Bangko Sentral ng Pilipinas is assumed to operate a managed float rather than a hard peg. The actual USDPHP rate is treated as reflecting market dynamics subject to intervention smoothing, not a policy-fixed value. A hard peg would render the DXY normalization trivially invalid.

3.7 Limitations

Four structural limitations constrain the interpretive reach of the findings.

The DXY basket is EUR-dominated, with the euro accounting for 57.6% of the index weight (ICE Benchmark Administration, 2023). The Philippine peso does not appear in the DXY basket. This means DXY movements conflate EUR/USD dynamics with broader dollar momentum. The correct normalization vehicle for the PHP would be a trade-weighted bilateral index using Philippine export and import partner currencies, weighted by trade volume shares. Such a daily-frequency series is not publicly available from the PSA or BSP. The DXY serves as the best available proxy, but it introduces sys-

tematic approximation error of unknown sign.

BSP FX intervention is omitted from both models. The BSP publishes gross international reserves at monthly frequency but does not disclose its daily intervention volumes in real time (Bangko Sentral ng Pilipinas, 2024). Reserve operations that suppress USDPHP upward pressure create a policy-induced component in the observed rate that neither the actual USDPHP nor the counterfactual captures. Both counterfactuals therefore represent lower bounds on the fundamental PHP rate. The true rate, absent both USD weakness and BSP intervention, would exceed the 64.93 DXY-normalized figure.

The ARIMA model has no causal structure. It cannot attribute peso movements to inflation differentials, current account dynamics, risk premium shifts, or capital flow reversals. The model projects a historical pattern forward without explanation. Meese and Rogoff (1983) demonstrated that such projections do not outperform a random walk in out-of-sample evaluation, which is precisely why confidence intervals widen rapidly beyond the short horizon. The ARIMA counterfactual should be read as a trend reference, not a causal forecast.

The OFW remittance channel is unmodeled. Remittance inflows generate persistent USD supply in the Philippine market that suppresses USDPHP independently of monetary policy or DXY dynamics (World Bank, 2023). The scale of this suppression, approximately US$36 billion annually, implies that the structural floor on peso depreciation is lower than either model suggests. Omitting this channel means the counterfactual rates overstate how much the peso would depreciate under a stable dollar, partially offsetting the intervention-omission bias described above.

4 Results

4.1 DXY Performance Since Inauguration

The DXY opened the Trump administration at 108.06 on January 20, 2025 and declined to 100.15 by March 27, 2026, a cumulative deterioration of 7.91 index points or 7.32%. Figure 1 documents this trajectory. The dollar weakened progressively from February 2025 onward, with the steepest decline coinciding with the announcement and escalation of broad tariff schedules in Q1–Q2 2025. The dollar did not recover to its inauguration-day level at any point in the study period. As of the analysis date, the DXY remained approximately 7.32% below the administration's opening value.

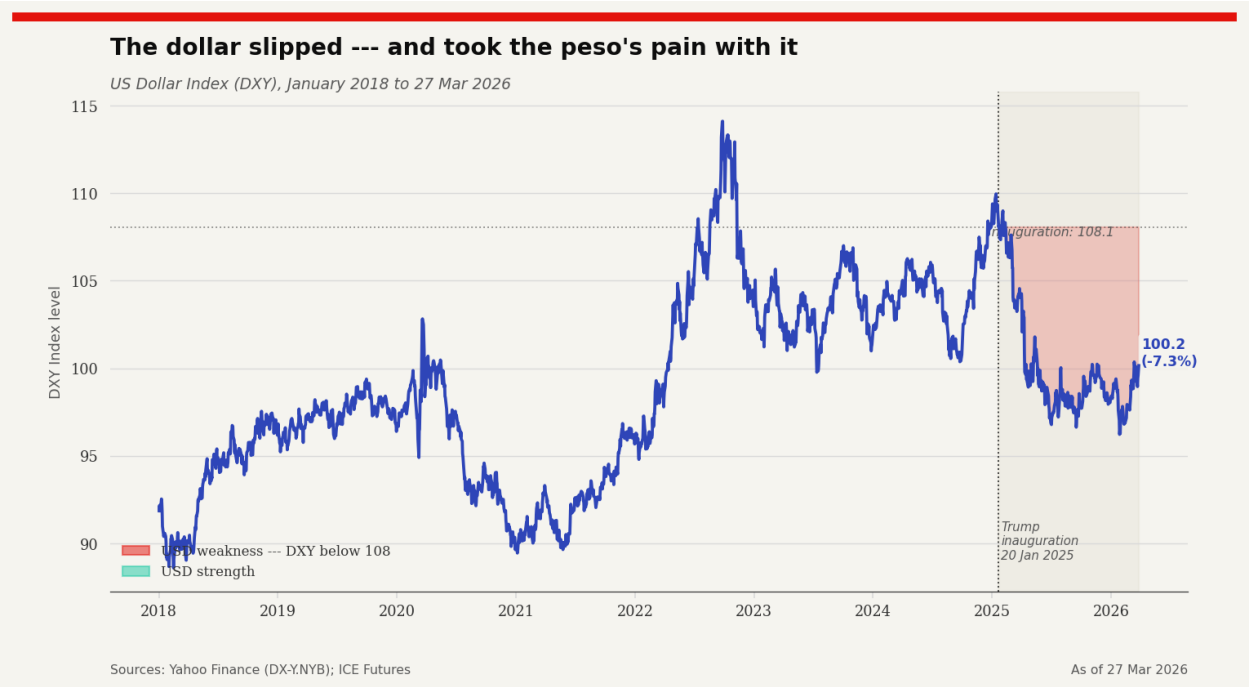

Figure 1
US Dollar Index from January 2018 to March 2026.

Note. The DXY declined from 108.06 to 100.15 between inauguration day (January 20, 2025) and March 27, 2026, a deterioration of 7.32%. The shaded red area marks the period during which the DXY remained below its inauguration-day baseline. Data sourced from Yahoo Finance via DX-Y.NYB.

4.2 Actual USDPHP Movement

The USDPHP spot rate rose from 58.15 on inauguration day to 60.21 on March 27, 2026, a nominal movement of 2.06 pesos per dollar over 14 months, representing a 3.54% nominal peso depreciation. Figure 2 presents the full 2018–2026 time series of USDPHP, overlaid with both counterfactuals.

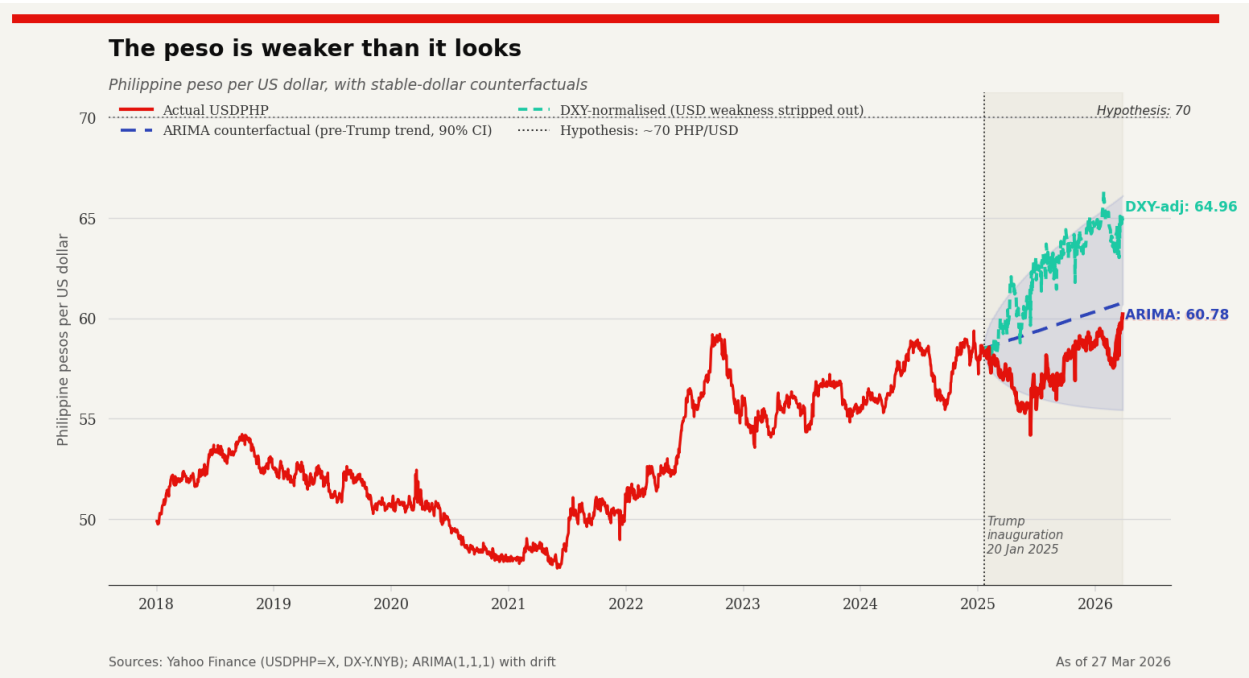

Figure 2
Philippine Peso Per US Dollar, January 2018 to March 2026.

Note. The solid red line depicts the actual USDPHP spot rate. The dashed blue line and shaded band depict the ARIMA(1,1,1) counterfactual with 90% confidence interval. The dash-dot teal line depicts the DXY-normalized counterfactual. The dotted grey line marks the 70 PHP/USD hypothesis. The vertical dotted line and shaded region mark the post-inauguration period. Data sourced from Yahoo Finance (USDPHP=X, DX-Y.NYB).

4.3 DXY-Normalized Counterfactual

Applying Equation 1 with $D_0 = 108.06$ and $D_t = 100.15$ for the March 27, 2026 observation:

$$\hat{e}_{\text{Mar 27, 2026}} = 60.21 \times \frac{108.06}{100.15} = 64.93 \text{ PHP/USD}$$

The DXY-adjusted rate of 64.93 implies a PHP depreciation of 6.78 PHP/USD relative to the inauguration baseline of 58.15, a fundamental deterioration of 11.66%. This is 8.12 percentage points larger than the nominal 3.54% depreciation the exchange board reports.

Figure 3 presents the post-inauguration period in detail, comparing the three series.

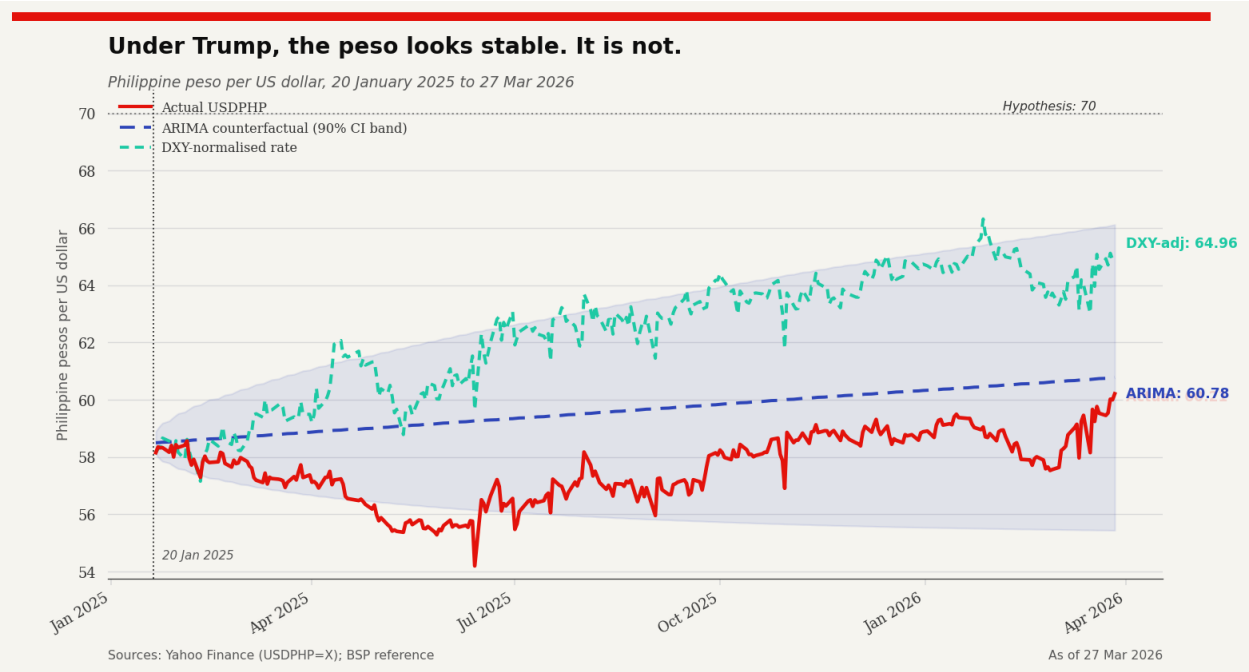

Figure 3
USDPHP Post-Inauguration, January 2025 to March 2026.

Note. All three series diverge progressively as DXY weakness widens. The actual rate (red, solid) trails the DXY-normalized rate (teal, dash-dot) by 4.72 PHP/USD at the terminal date. The ARIMA counterfactual (blue, dashed) closely tracks the actual rate, confirming the peso is depreciating near its pre-Trump trend in nominal terms. Data sourced from Yahoo Finance.

4.4 PHP Cushion Trajectory

The PHP cushion Γ_t, defined in Equation 2, began near zero at inauguration and widened progressively as the DXY declined. By March 27, 2026, Γ_t reached 4.72 PHP/USD. Figure 4 displays the full trajectory. The cushion was positive throughout the post-inauguration period, meaning the dollar's weakness consistently masked Philippine-specific depreciation pressure.

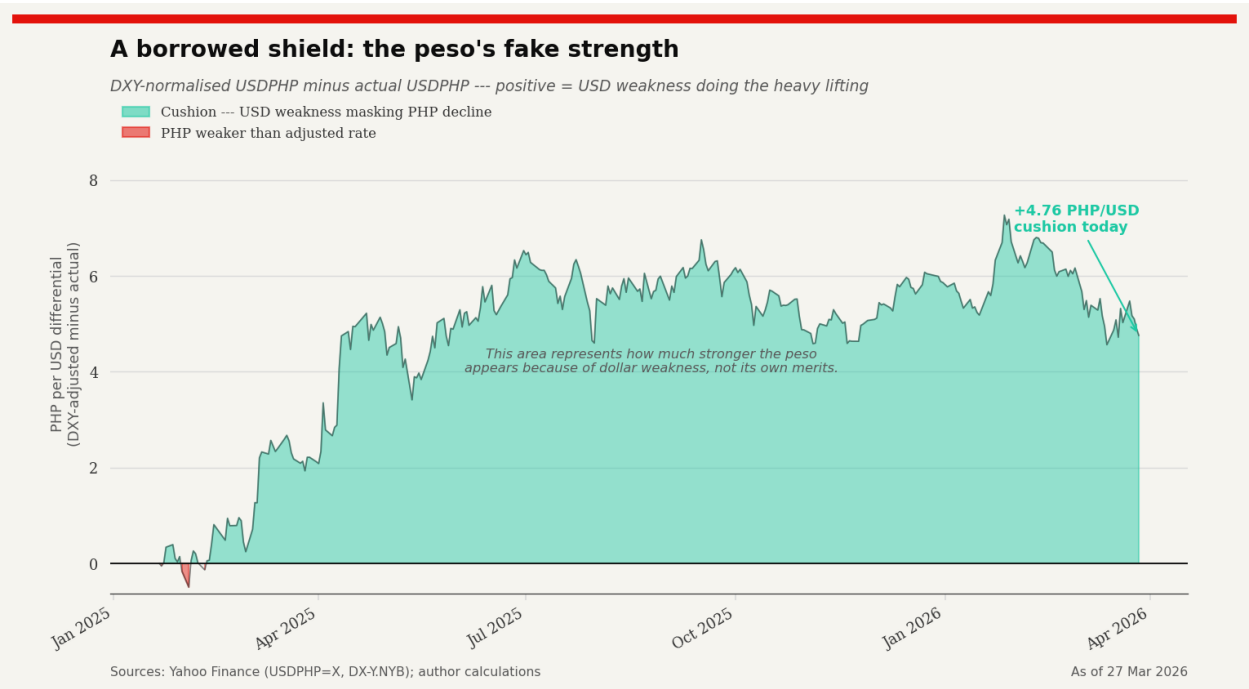

Figure 4
PHP Cushion: DXY-Normalized Rate Minus Actual Rate.

Note. A positive value indicates that USD weakness is suppressing the observed USDPHP rate below its DXY-adjusted counterfactual. The cushion reached 4.72 PHP/USD by March 27, 2026. The shaded teal area represents the cumulative benefit the peso has received from dollar deterioration. Author calculations based on Yahoo Finance data.

4.5 ARIMA Forecast Results

The ARIMA(1,1,1) model produced a point forecast of 60.78 PHP/USD for March 27, 2026, with a 90% confidence interval of approximately [58.90, 62.70]. The actual rate

of 60.21 lies within this interval and is 0.57 pesos below the point forecast. This result indicates that the peso is depreciating at approximately the rate its pre-Trump trend predicted in nominal terms. The ARIMA forecast does not capture the fundamental rate, however: it projects the nominal path, which incorporates the dollar's concurrent weakness.

4.6 Evaluation of the 70 PHP/USD Hypothesis

The hypothesis that the USDPHP rate would sit near 70 under a stable-dollar scenario is directionally correct. Both counterfactuals confirm that the official rate materially understates PHP weakness. However, the DXY-adjusted rate of 64.93 falls short of 70 by 5.07 PHP/USD, or 7.25%. Reaching 70 would require the DXY to have fallen from 108.06 to approximately 93 to 95, a decline of 13 to 14%, compared to the observed 7.32%. Table 1 presents the full set of summary statistics.

Table 1
Summary Statistics, Philippine Peso Depreciation Analysis.

Key metrics for the USDPHP rate and DXY from January 20, 2025 to March 27, 2026.

Metric	Value	Interpretation
DXY at inauguration (Jan 20, 2025)	108.06	USD strength baseline
DXY at analysis date (Mar 27, 2026)	100.15	−7.32% from baseline
USDPHP at inauguration	58.1500	PHP reference rate
USDPHP at analysis date (actual)	60.2060	+3.54% depreciation
USDPHP DXY-normalized counterfactual	64.9333	+11.66% implied depreciation
USDPHP ARIMA forecast	60.7750	+4.51% trend projection
PHP cushion (Γ_t) at analysis date	+4.72 PHP/USD	USD weakness masking peso
Hypothesis (stable-dollar rate)	≈70 PHP/USD	Overstated by ≈5 PHP/USD

Note. DXY data from Yahoo Finance (DX-Y.NYB). USDPHP data from Yahoo Finance (USDPHP=X). The DXY-normalized counterfactual applies Equation 1. The ARIMA forecast is the point estimate from an ARIMA$(1, 1, 1)$ model with drift fitted on July 2020 to January 2025 training data. The PHP cushion is defined per Equation 2.

5 Analysis and Discussion

5.1 The Decomposition Finding and Its Implications

The central finding is a divergence of 8.12 percentage points between nominal and fundamental PHP depreciation over 14 months. The actual rate declined 3.54%. The DXY-adjusted rate declined 11.66%. The gap is the PHP cushion, and it has grown monotonically since inauguration day.

This divergence carries direct consequences for economic agents. Philippine importers pricing in USD face an effective cost increase of 11.66% relative to the peso- denominated prices their balance sheets report. The official USDPHP rate suggests an 11.66%

exposure has cost them 3.54% – a comfortable underestimation of true import cost inflation. Exporters receiving dollar revenues face the mirror-image problem: fewer pesos per exported dollar than fundamentals would imply under a stable-dollar baseline, which the OFW remittance channel amplifies at scale.

The BSP's managed float regime interacts with this dynamic in a compounding way. Calvo and Reinhart (2002) document that fear-of-floating central banks suppress exchange rate volatility through reserve operations. If the BSP is simultaneously suppressing upward USDPHP pressure through FX intervention, then the 4.72 PHP/USD cushion identified here understates the fundamental gap. The true policy-adjusted counterfactual rate would be higher than 64.93.

5.2 Why the Hypothesis of 70 PHP/USD Falls Short

The directional logic of the 70 PHP/USD hypothesis is sound: a stronger dollar combined with the PHP's secular depreciation trend would have placed the rate materially above current levels. The quantitative error is one of magnitude.

The DXY declined 7.32%, producing a 4.72 PHP/USD cushion on a base rate of 58.15. To produce a 70 PHP/USD counterfactual, the required DXY decline would need to reach approximately 13 to 14%. That would represent a dollar weakening event of historic severity – exceeding the declines observed during the 2008 financial crisis and the 2020 pandemic shock. The Trump administration's tariff policies and fiscal uncertainty have been disruptive, but they have not produced a DXY collapse of that magnitude within the study window. A 70 PHP/USD rate remains consistent with a more extreme dollar deterioration scenario or a longer accumulation period.

5.3 Model Limitations in Context

The four structural limitations documented in the Methodology section bear directly on the interpretation of results. Two competing biases operate in opposite direc-

tions. The omission of BSP intervention causes both counterfactuals to understate the true fundamental rate, because active reserve operations suppress USDPHP below what market forces alone would produce. The omission of OFW remittance inflows partially offsets this, because structural dollar supply from remittances sets a natural floor on peso depreciation. The net direction of these competing omissions is indeterminate without additional data.

The DXY basket mismatch introduces a second-order bias of unknown sign. If EUR/USD dynamics over the study period overstated the dollar's effective weakness against Philippine trade-partner currencies, the 64.93 counterfactual is too high. If they understated it, the counterfactual is too low. Access to a Philippines-specific trade-weighted dollar index would resolve this ambiguity.

5.4 Structural Peso Weakness and the Remittance Channel

The OFW remittance channel is simultaneously a stabilizer and an analytical confound. World Bank (2023) estimates that the Philippines received approximately US$36 billion in remittances in 2022, among the highest in Southeast Asia as a share of GDP. During periods of dollar weakness, each dollar remitted generates fewer pesos. The households receiving these remittances experience an effective income decline in peso terms even when the quoted exchange rate appears stable.

This creates a policy puzzle for the BSP: a stable USDPHP rate during a period of dollar weakness provides no protection to remittance-dependent households. The exchange board shows stability; the welfare impact is contraction. The decomposition in this paper makes this discrepancy explicit.

5.5 The Rogoff Puzzle and Persistence

Rogoff (1996) demonstrated that deviations from purchasing power parity equilibrium dissolve slowly, with half-lives of three to five years. The PHP cushion created by

DXY weakness is therefore unlikely to correct in a single quarter even if the DXY fully recovers. The adjustment path depends on BSP policy responses, remittance flows, Philippine inflation relative to US inflation, and external capital account conditions. The persistence forecast from Rogoff's framework implies that Philippine economic planners should not assume the cushion will vanish quickly following any DXY recovery.

The numbers behind the peso's borrowed calm

Key metrics, Philippine peso vs US dollar --- as of 27 Mar 2026

Metric	Value	Interpretation
DXY at inauguration	**108.06**	*USD strength baseline*
DXY today	**100.15**	*-7.3% --- dollar weakened*
USDPHP at inauguration	**58.1500**	*PHP reference rate, 20 Jan 2025*
USDPHP today (actual)	**60.2060**	*+3.54% vs inauguration*
USDPHP DXY-normalised	**64.9612**	*+11.71% implied depreciation*
USDPHP ARIMA forecast	**60.7750**	*+4.51% pre-Trump trend*
User hypothesis (stable USD)	**~70.0**	*Overshoots by ~5 PHP/USD*
PHP cushion from USD weakness	**+4.7552 PHP/USD**	*Dollar drop masking PHP losses*

Sources: Yahoo Finance (USDPHP=X, DX-Y.NYB); BSP; ARIMA(1,1,1) model

As of 27 Mar 2026

Figure 5
Summary Statistics Table, Philippine Peso Depreciation.

Note. Complete numerical summary of all key metrics computed in this analysis. Data sourced from Yahoo Finance (USDPHP=X, DX-Y.NYB). Counterfactuals are author calculations using the methods defined in Equations 1 and 3.

6 Conclusion

This paper established that the Philippine peso is functionally weaker than exchange boards report. The nominal USDPHP depreciation of 3.54% between January 20, 2025 and March 27, 2026 substantially understates the peso's fundamental position. After stripping out the 7.32% decline in the US Dollar Index using the DXY-normalization framework in Equation 1, the fundamental PHP depreciation is 11.66% over the same period, a gap of 8.12 percentage points attributable entirely to dollar weakness.

The ARIMA(1,1,1) counterfactual, projecting the pre-Trump trend forward, yields a terminal forecast of 60.78 PHP/USD, closely tracking the nominal rate and confirming that the peso's nominal path is broadly consistent with its pre-administration depreciation trend. The nominal stability is real; the fundamental stability is borrowed.

The PHP cushion of 4.72 PHP/USD at the analysis date represents the peso's reliance on exogenous dollar weakness for its apparent resilience. This cushion grew monotonically across the study period as the DXY declined. Its unwinding, when it occurs, will expose the peso to a correction that the nominal rate has been deferring.

The 70 PHP/USD stable-dollar hypothesis is directionally correct and analytically defensible. Its quantitative execution requires a DXY decline of approximately 13 to 14%, not the observed 7.32%. The counterfactual rate under actual DXY conditions is 64.93, not 70. The difference of 5 PHP/USD is not a failure of reasoning but a calibration of magnitude.

Three implications follow from these findings. First, Philippine importers and financial planners should price their USD exposures against the DXY-adjusted rate of 64.93, not the official 60.21. Second, BSP policy statements citing peso stability should be read against this decomposition: the stability is partly dollar-provided. Third, OFW remittance planning should account for the effective income erosion that dollar weakness creates in peso-denominated terms, a welfare effect invisible at the quoted exchange rate.

Future research should extend this framework in two directions. A bilateral, trade-weighted normalization using Philippine trade partner currencies would improve on the DXY proxy used here. A structural model incorporating BSP intervention data, Philippine inflation differentials, and OFW remittance volumes would permit a more causally precise decomposition of the PHP cushion into its policy and market components.

The peso's calm is borrowed. The bill will arrive.

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